

Comparative flavonoids contents of selected herbs and associations of their radical scavenging activity with antiproliferative actions in V79-4 cells.

To elucidate the health benefit of herbal teas on the cytotoxicity induced by H_2O_2 in V79-4 cells, herbal extracts and its flavonoids were tested using lactate dehydrogenase release and determining intracellular reactive oxygen species generation and antioxidant activity with superoxide radical scavenging assay. Significant decrease in cell viability was observed on V79-4 cells treated with H_2O_2 (1 mM), while herbal extracts and its flavonoids including catechin and epigallocatechin gallate prevented the LDH release from H_2O_2 cytotoxicity. Total catechin contents of green tea (65.6 mg/g of dry matter) were significantly higher than other herbal teas (35.8 to 1.2 mg/g of DM). The relative concentration of the 4 major tea catechins ranked EGCG > EGC > EC > C. Green tea exhibited the lowest IC(50) values (2 g fresh herb/100 mL) of superoxide radical scavenging activity among the tested herbal tea, which indicates powerful antioxidant activity in $O_2(^{\cdot-})$ radicals scavenging, followed by black tea, dandelion, hawthorn, rose hip, chamomile.